

LoveSims: Exploring ‘What-If’ Scenarios for Relationship Insights and Compatibility

Mateo Larrea*
Stanford University
Stanford, California, USA
mlarrea@stanford.edu

Xingyi Zhang*
Stanford University
Stanford, California, USA
xingyiz@stanford.edu

Xuyang Zhu*
Stanford University
Stanford, California, USA
xuyang1@stanford.edu

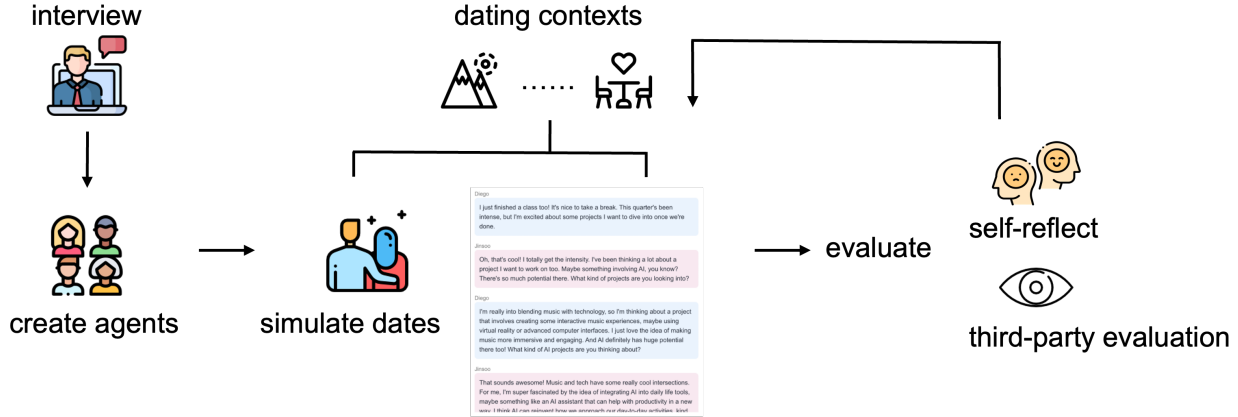


Figure 1: The *LoveSims* framework involves creating agents that simulate human behavior during dates through interviews, running date simulations, and conducting iterative evaluations through self-reflection (by agents and by users) and third-party evaluation (LLMs).

Abstract

Existing dating technologies often prioritize compatibility algorithms and filtering mechanisms but fail to address the complexities of long-term relationship development. As a result, users are confined to the early stages of dating, with limited tools to navigate the deeper dynamics required to build lasting connections. This paper introduces *LoveSims*, a generative agent-based simulation framework, which establishes a theoretical foundation for modeling the potential evolution of romantic relationships. *LoveSims* simulates structured interactions through personalized agent personas, enabling users to explore “what-if” scenarios and gain insight into communication patterns, compatibility, and potential relationship challenges. Our evaluation compared the alignment between synthetic agent responses and real-world speed-dating data. The findings demonstrate that agent preferences closely mirror patterns observed in human studies. Moreover, self-reflection by agents provides more accurate and consistent insights than third-party evaluations.

* All authors contributed equally to this research.

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CCS Concepts

• Computing methodologies → Agent / discrete models; Multi-agent systems.

Keywords

Generative Agents, Simulation, Social Relationships, Dating

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1 Introduction

Social computing systems play a significant role in mediating dyadic relationships, particularly in the context of romantic connections [6, 18]. Dating technologies exemplify this trend, especially in their focus on early-stage interactions like matchmaking and swiping. However, existing methodology rely on algorithms to calculate compatibility scores based on user-provided data, but these methods often reinforce existing biases and fail to capture the nuanced complexities of human relationships [16]. During the swiping and initial contact phase, many apps employ a constant filtering approach aimed at eliminating unsuitable matches rather than identifying promising connections. This design fosters a “shopping culture” in

romantic relationship formation, reducing the process of interpersonal connection to one of feature-matching [1]. Such an approach undermines the development of deeper, meaningful relationships.

In this paper, we use romantic relationship as an example to inform the broader literature space of dyadic relationship formations in social computing. To foster more effective relationship mediation, technology must move beyond matchmaking and refocus on facilitating genuine interpersonal interactions. Current dating technologies face two primary limitations. First, they require users to navigate early-stage interactions through repetitive trial-and-error processes, which can be both time-consuming and emotionally draining [7]. Second, they fail to account for long-term relationship development concerns, such as potential conflicts, evolving dynamics, and individual factors that influence long-term compatibility [2]. This information gap between dating partners can negatively impact relationship development. Advancing technology in this domain should address these issues by prioritizing users at the center of the relationship development process.

Recent advancements in social simulations with generative agents have shown increasingly promising outcomes in simulating human behaviors across various social contexts, including online communities [13], small towns [12], and workplace scenarios (e.g., hospitals [4]). The common architecture of generative agents—comprising an agent profile, memory, and subsequent planning and actions [19]—has proven capable of producing believable agent behaviors. Researchers have also explored the potential of generative agents to simulate real people for conducting social science research, such as modeling populations of 1,000 individuals [14]. While applying these agent architectures to simulate romantic relationship development is appealing, a significant challenge lies in the absence of clearly defined states for long-term interactions aimed at specific goals, such as progressing from the first conversation to entering a romantic relationship with an ideal match. Current systems also lack mechanisms for synchronous feedback, which would allow agents to become more representative of their users over time.

In this paper, we introduce *LoveSims*, a generative agent simulation framework that redefines the role of technology in the development of romantic relationships. *LoveSims* shifts the focus from superficial matchmaking to cultivating meaningful relationship insights and growth through simulations under different contexts and evaluation mechanisms. Our goal is to alleviate the burden of initial interactions and provide users with helpful simulation outcomes that close the information gap between dating partners by predicting future relational dynamics and highlighting potential challenges in their relationships. The system incorporates two key innovations: (1) including a variety of contexts for long-term interactions and (2) integrating a feedback loop to enhance the fidelity of agent simulations based on user input. Together, these features enable a shift from superficial matchmaking to a more human-centered approach focused on relationship insights and growth.

We also conducted a preliminary evaluation study to validate the alignment of *LoveSims* agents with human data. To address the evaluation question—*How accurately can generative agents simulate the formation of romantic relationships, and what factors contribute to creating a personalized ideal partner?*—we developed 20 synthetic generative agents and observed their interactions during simulated

dates. The results were evaluated using key metrics informed by established behavioral science methods, including those commonly employed in speed-dating studies.

2 Related Works

In this section, we review the prior literature on generative agent-based simulations and the trend of simulating human behaviors. We also reflect on previous behavioral science research on human dating preferences, which provides foundational methods for validating the alignment of generative agents in dating interactions.

2.1 Generative Agents and Social Simulations

Generative agents have been increasingly used to model human-like behavior in various contexts, including virtual communities [13], workplace simulations [20], and large-scale social experiments [11]. These systems often combine agent profiles, memory systems, and action-planning mechanisms to simulate interactions. For example, Park et al. [12] utilized generative agents to replicate societal trends for social science research, while Zhou et al. [20] simulated workplace environments to study organizational dynamics. However, many current implementations lack mechanisms for modeling emotional intelligence or long-term relational dynamics, which are essential for simulating romantic relationships. Recent work has introduced memory architectures that enhance contextual adaptation. Park et al. proposed a memory stream system that enables agents to recall and adapt based on past interactions [12]. Similarly, Croissant et al. [3] developed an appraisal-based emotional model for game agents, allowing for dynamic emotional responses. Despite these advances, the integration of emotional states with memory systems in generative agents for relational modeling remains underexplored.

2.2 Speed Dating Experiments

Speed dating research offers valuable insights into romantic relationship and serves as a benchmark for evaluating simulated social interaction in creative contexts. Studies have identified key factors influencing attraction, such as physical appearance, shared interests, and perceived similarity [5, 8, 10]. Additionally, perceived similarity often outweighs actual similarity in predicting attraction [17]. Comparison to these findings with generative agent simulations provides more insights on realistic modeling of romantic dynamics.

LoveSims builds on these insights by leveraging generative agents to simulate romantic relationships. The iterative self-reflection mechanisms and context-based interactions enable agents to incorporate memories of prior dates and inform users about potential future dating dynamics. By aligning simulation outcomes with patterns observed in human behavioral studies, *LoveSims* established the theoretical foundation for simulating romantic relationships with generative agents.

3 Methodology

3.1 Agent Creation

Synthetic Agents (Implemented Approach): We utilized synthetic agents from Park et al.’s dataset [14], which employs demographic diversity to simulate varied social interactions. We randomly selected 20 agents from this dataset, which consists of 1,052 generative agents representing real individuals recruited through stratified sampling based on age, census division, education, ethnicity, gender, income, neighborhood, political ideology, and sexual identity. Using this framework, we generated detailed life stories and memories via ChatGPT, designing agents that exhibit realistic behaviors, personalities, and decision-making while maintaining computational efficiency.

Real-Person Agents (Proposed Approach): In addition to synthetic agents, we propose creating agents modeled on real individuals. The proposed method involves structured interviews and surveys to collect biographical, behavioral, and value-based data. The interviews may cover personal narratives (e.g. life stories, aspirations), educational and social life, rapid-fire questions (e.g. instinctive responses and personality traits) and personal values.

3.2 Simulation Structure

The **LoveSims** program creates a simulated romantic date scenario, where agents interact within a predefined context for a set number of message exchanges. Users can choose a primary agent (e.g., Diego) to engage in one of two configurations:

- (1) **One-to-One:** A single agent engages in a simulated date with another chosen agent.
- (2) **One-to-All:** The primary agent interacts sequentially with every agent in the agent bank.

Users can customize each simulation by specifying both the context and duration of the interaction. For example, contexts might include scenarios like, “You are on your first date at Treehouse Cafe during the last day of the fall quarter,” or, “You are meeting for a casual coffee at a local cafe on a rainy afternoon.” Interaction durations can be adjusted by setting the number of messages, allowing for varying levels of conversational depth. Users can also simulate future scenarios by tailoring the context to reflect hypothetical or planned situations, such as, “You’re on your 10th date and discussing the possibility of your partner meeting your parents.” The agents adapt their behavior accordingly, enabling users to explore potential dynamics and outcomes in advance.

During each interaction, the program utilizes OpenAI’s GPT-4 model via API calls to generate realistic, persona-driven responses (Figure 2). Advanced prompt engineering techniques were employed to ensure that agents role-played their assigned personas, maintaining consistency in their traits, preferences, and conversational styles. This combination of context-specific customization and tailored AI responses created highly immersive and engaging simulations.

3.3 Self-Reflection and Third-Party Evaluation

At the conclusion of each interaction, agents conduct a self-reflection exercise to evaluate their dating experience, especially how they feel about the other agent (see Figure 3). **Self-reflections** leverage

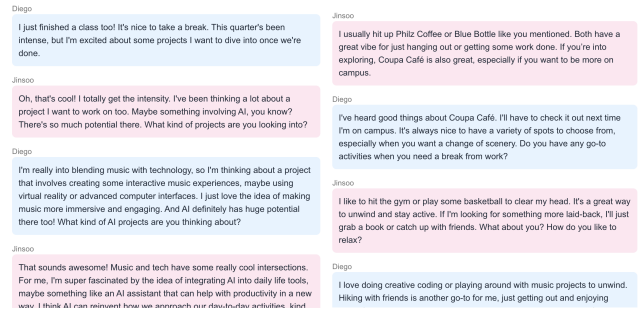


Figure 2: Chat interface showing a casual conversation between two agents.

agent profile and defined character to evaluate transcripts of the simulated dates. This allows the agents to assess factors such as conversational flow, compatibility, and emotional engagement by themselves. For instance, Leila reflects on her interaction with Yuki, noting, “The conversation felt warm and genuine, and I appreciated their calm demeanor. While there wasn’t an overwhelming spark, there was a comfortable connection.”

In parallel, a non-role-playing large language model (LLM) performs a **third-party** evaluation, providing an external analysis of the interaction. These evaluations focus on objective metrics like conversational depth, shared interests, and overall rapport. For example, the evaluation of Leila and Yuki’s date highlights, “while the conversation flowed smoothly, it remained superficial, lacking deeper shared values or emotional bonding.”

This dual evaluation method provides a comprehensive view include both subjective perceptions and objective compatibility patterns in one interaction instance. Additionally, by running simulations across multiple contexts, we examined how environmental factors influenced relational dynamics, uncovering nuanced patterns of agent behavior and preferences.

More importantly, this reflection process was designed not only to encourage refinement of the agent’s persona but also to encourage the user to observe their agent’s behavior more closely. By examining interactions and evaluations, the simulations highlighted patterns or tendencies that allowed the user to better understand dynamics and potential outcomes. This feedback loop served as a deliberate mechanism for improving agent accuracy while providing a lens through which users could gain insight into relational and conversational behaviors.

3.4 Data Collection and Analysis

Data collection involved running the one-to-all configuration for 20 synthetic agents across three different contexts (café, museum, restaurant). Each context was tested four times to account for variability, resulting in a total of 4560 simulated interactions (380 pairs $\times$ 3 contexts $\times$ 4 runs). We calculated the *positive response rate* as how many positive responses an agent gets. For example, if an agent met with 10 agents and 6 of them indicated that they would like to meet again, the positive response rate would be 60%. We also calculated *attribute scores*, which were the averaged scores the

Run Evaluation

Choose how to evaluate the simulation results

☒ Self Reflection
Agent reflects on the interaction using post-simulation questions

☐ Third-Party (Transcript)
Analyze compatibility based on conversation transcript

☐ Third-Party (Profiles)
Analyze compatibility based on agent profiles and memories

☐ Run All Evaluations
Perform all applicable evaluation types

Run Evaluation

Evaluation Results

Diego's Self-Reflection Analysis

Analysis:
The interaction with Jinsoo was pleasant and engaging. We talked about mutual interests like cafes and outdoor activities, which suggests some compatibility. There wasn't a strong romantic spark, but no red flags or deal-breakers emerged from the conversation. Jinsoo seemed sincere and easy to talk to, and I appreciated the positivity and common ground in our interests. However, the conversation largely stayed on surface-level topics and didn't dive deeper into personal values or unique perspectives, which is something I'd want in a stronger connection. Overall, it was a good interaction, worth exploring further to see if a deeper connection might develop.

Decision:
Would Diego see them again? **yes**

Satisfaction Score: 65/100
Length Feedback: Just right

Attribute Ratings:

- Ambition: 60/100
- Attractiveness: 70/100
- Fun: 60/100
- Intelligence: 85/100
- SharedInterests: 72/100
- Sincerity: 75/100

Figure 3: Agent's Self-Reflection evaluation after a date.

agent received from the 19 other agents. We collect both subjective (self-reflection metrics) and objective measures (third-party metrics), which each include: (a) Attribute scores on ambition, attractiveness, fun, intelligence, shared interests, and sincerity; (b) Decision on future engagement.

4 Results

4.1 Comparison to Human Experiments

We measured the correlation between attribute scores and positive response rates (see Figure 4 for examples). In the self-reflection condition, the attribute scores and positive responses were provided by the agents themselves. In contrast, in the third-party condition, a third-party LLM evaluated the transcripts and assigned the scores. These correlations (R values) between attributes and positive response rates reveal which attributes individuals prioritize in mating decisions.

The third-party LLM evaluates interactions using structured prompts designed to assess satisfaction, compatibility, and attribute similarity (see Appendix for full prompt specifications). It assigns scores based on specific criteria, such as conversational flow, emotional connection, and shared interests. These assessments derive from a structured analysis process inspired by decision-making frameworks used in psychological studies [5].

The metrics in Figure 5 were computed by aggregating these LLM-generated scores across multiple trials and contexts. Given the nature of the LLM's evaluation, one potential confounding factor is agreeableness bias, where agents that are generally more positive or socially engaging may receive higher scores. This could reduce variance in attribute-response correlations, particularly if the model interprets politeness or enthusiasm as indicators of compatibility. To account for this, we analyzed how frequently agents received high scores across different contexts, ensuring that scoring trends were not solely driven by an agent's general positivity but rather by context-specific interactions.

We aggregated the R values across all three contexts (Figure 5). The overall pattern of prioritized attributes (attractiveness, shared interests, and fun) aligns with findings from real-person experiments [8]. However, an excessively high positive response rate might reduce these correlations in simulations, making them lower compared to the real-person experiments. Notably, the self-reflection condition outperformed the third-party condition, demonstrating the effectiveness of role-playing. One exception was the sincerity attribute, which showed a much stronger correlation—likely

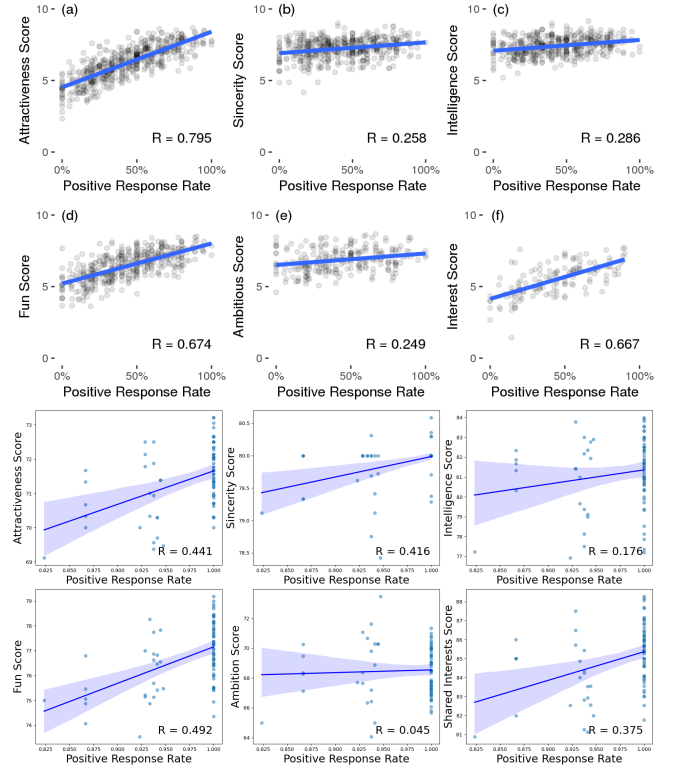


Figure 4: Distributions of positive response rates versus attribute scores. Top: Results from real-person experiments [8]. Bottom: Results from simulations in the museum context, based on self-reflection metrics.

because evaluating sincerity between two agents can be inherently ambiguous.

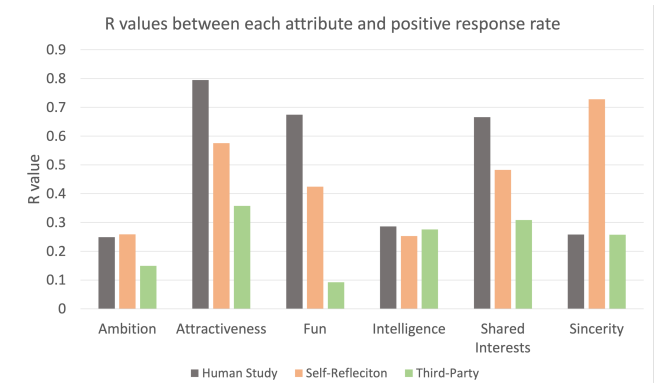


Figure 5: Correlations between positive response rates versus attribute scores in human study[8], self-reflection based metrics, and third-party based metrics in simulation.

4.2 Consistency

We assessed consistency using two measures. The first, response consistency, evaluates whether pairs give consistent responses across all four trials in the same context. Yet this measure doesn’t distinguish between the ‘2-2’ case (2 yes, 2 no) and the ‘3-1’ case, although the former is more inconsistent. The second measure, majority response rate, captures how strongly one response dominates. For example, a 2-2 case results in a 50% majority response rate, a 3-1 case yields 75%, and unanimous responses lead to 100%.

Under both measures, the simulations have high overall consistency, with the Majority Response Rates above 90% for both metrics types (Table 1). The self-reflection-based metrics showing greater consistency than third-party metrics.

4.3 Context

Contexts impact results. A pair of agents that match in one setting may not match in another. Differences between real-person experiments and certain simulated contexts do not invalidate the results. For example, real data from speed-dating experiments may most closely align with the café context. Variations across contexts suggest the possibility of higher ecological validity, highlighting the program’s potential to simulate scenarios beyond what has been observed in real-world experiments.

5 Discussion and Future Work

5.1 Current Limitations

Our study highlights several limitations in the current implementation of *LoveSims*, which serve as opportunities for future enhancement. Firstly, the generative agents exhibit an intrinsic bias towards politeness and agreeability, a characteristic inherited from large language models (LLMs). This results in an overrepresentation of positive responses, such as agents frequently agreeing to meet their partners again, which may not accurately reflect real-world dynamics. While this aligns with LLMs’ tendency to avoid conflict, it diminishes the diversity and authenticity of interactions. It is also worth noting that such a tendency to match only exists in the self-reflection condition but not in the third-party condition.

Beyond this bias, LLMs struggle with capturing the full spectrum of human social behavior, particularly in complex psycho-social and psycho-sexual interactions. Romantic decision-making is shaped by non-verbal cues such as tone, body language, and microexpressions—elements that LLMs, operating purely through text, cannot perceive or generate. As a result, interactions remain limited to explicit verbal exchanges, lacking the implicit emotional depth that often guides real-world romantic attraction and social compatibility.

LLM-generated agents struggle with emotional fluctuation and unpredictability. Human relationships involve misunderstandings, shifting moods, and evolving desires, but LLMs rely on pre-trained conversational patterns, making responses feel static or overly rational. Without personal experience, they fail to capture idiosyncratic behaviors, attachment styles, and deep-seated biases that shape romantic decision-making.

The distinction between self-reflection and third-party evaluations further highlights inconsistencies. Self-reflective agents reinforce positive outcomes, while third-party evaluations tend to

be more detached. This raises concerns about the reliability of LLM-based assessments in ambiguous social scenarios.

5.2 Future Work

Building upon these insights, several avenues for future research and development emerge. Enhancing prompt engineering to introduce greater contextual and emotional variability is a critical next step. A more sophisticated prompt design could simulate a broader range of relational dynamics, enabling agents to respond to complex scenarios with greater realism.

Improving the interview and survey process is another priority. By accounting for emotional contexts and refining the data collection methodology, we can create richer, more accurate agent personas. This would enhance the agents’ ability to reflect the diversity of human experiences and relational preferences.

A significant improvement would involve integrating an architecture for affective states, informed by past psychology research and feedback from experts (e.g., psychologists or relationship counselors). Leveraging advancements in emotional simulation, such as the appraisal-based models proposed in [15] could allow agents to simulate emotional responses, such as empathy or frustration, thereby creating more engaging and realistic interactions. Additionally, adopting elements from affective computing, including emotion recognition and context-aware response generation, could enhance the agents’ ability to mirror human emotional dynamics. Our study also highlights the potential of AI-driven simulations for relationship counseling and dating guidance, yet we recognize that further expertise and refinement are necessary for these tools to be truly effective.

Improving the memory stream architecture with a retrieval function (e.g. proposed in [12]) would enable agents to better retain and recall past interactions. This would facilitate more dynamic and evolving simulations, where agents could build upon shared histories and adapt their behaviors over time. Such an enhancement would contribute to a more authentic portrayal of relational growth and complexity.

Finally, validating our approach through real-world user studies is an essential next step. While our current work establishes a proof-of-concept within a controlled environment, conducting empirical studies with actual users will provide deeper insights into the system’s effectiveness and limitations. We can iteratively refine the model based on authentic behavioral patterns and feedback. For example, incorporating users’ synchronous feedback on both the level of fine details (e.g. what do they like/dislike about the conversation) and main result (e.g. whether they decide to match or not) can help the agents learn to represent the user’s intention. Such interaction-based improvement has also been explored in recent studies such as [9] where they allow therapists to interact with AI patients to create more authentic conversation.

Furthermore, ensuring the system’s generalizability across different cultural contexts is crucial for broader applicability. Romantic and social norms vary significantly across cultures, influencing communication styles, expectations, and attraction cues. Future work will incorporate culturally diverse datasets and interaction patterns to enhance the adaptability of *LoveSims*, making it more

Table 1: Consistency

Metrics Type	Consistency Rate (%)	Majority Response Rate (%)
Self-reflection	89.29	96.76
Third-party	75.62	92.38

inclusive and reflective of global relationship dynamics. These validations will help bridge the gap between simulation and reality, making *LoveSims* a more meaningful tool for studying and supporting romantic decision-making.

5.3 Ethical Considerations

As we advance this research, we must remain mindful of the ethical implications of our design choices. While incorporating sensitive data, such as race and ethnicity, could enhance the realism of simulations, it risks reinforcing stereotypes or biases. Future work must address these concerns through rigorous ethical frameworks, ensuring that our models promote inclusivity and fairness.

By addressing these limitations and expanding the capabilities of *LoveSims*, we aim to create a more sophisticated tool for exploring the dynamics of romantic relationships. These advancements hold promise not only for improving dating simulations but also for contributing to broader research in human-AI interaction and social modeling.

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A Prompt Templates for Evaluations in LoveSims

A.1 Self-Reflection Prompt

You are {agent_name}. Based on your profile:
{agent_profile}

And your memories:
{memories}

You just had a conversation with {other_agent} in the context: {context}

Here's the transcript:
{transcript}

Please reflect on this interaction by providing:

1. A satisfaction score (0-100). Be BRUTALLY honest.
2. Feedback on conversation length and engagement quality.
3. Ratings (0-100) for different attributes using the FULL range.
4. A critical analysis highlighting both positives AND negatives.

Don't sugarcoat anything. If something was bad, say it was bad.

A.2 Transcript Analysis Prompt

As a harsh critic, analyze this conversation between {agents}:

Context: {context}

Transcript:
{transcript}

Provide a brutally honest analysis:

1. Overall compatibility (0-100)
2. Score specific aspects (conversation flow, emotional connection, shared interests, red flags).
3. Key factors affecting compatibility.
4. Detailed critique of their interaction - don't hold

back on criticism.

A.3 Profile Analysis Prompt

As a ruthlessly honest matchmaker, analyze the compatibility between:

Person 1 ({agent1}):
Profile: {profile1}
Memories: {memories1}

Person 2 ({agent2}):
Profile: {profile2}
Memories: {memories2}

Provide a harsh assessment:

1. Overall compatibility potential (0-100).
2. Score values alignment, lifestyle compatibility, personality match, long-term potential.
3. Identify dealbreakers and red flags.
4. Detailed compatibility analysis - focus on potential problems.

Be merciless in your assessment. Most matches are NOT highly compatible.